ROAD ANOMALY DETECTION

Team:DeepVision

MEMBRES:

BILEL BEYNI

MOLKA GHARBI

EYA KASSOUS

MARIEM CHOUCHENE

CHAIMA CHERIF

MOHAMED AKRAM BOUZAIANE

Supervised by:Ms.Malek Hedhli

2026-2027

Contents

General Introduction **1**

1 Project Overview and Methodology **2**
1.1 Introduction . . . 2
1.2 The CRISP-DM framework . . . 2
1.3 Conclusion . . . 3

2 Business Understanding **4**
2.1 Introduction . . . 4
2.2 Problem definition . . . 4
2.3 Global objective . . . 4
2.4 Business objectives and Data Science objectives . . . 4
2.5 Project objectives . . . 5
2.6 Expected outcomes and success criteria . . . 6
2.7 Conclusion . . . 6

3 Data Understanding **7**
3.1 Introduction . . . 7
3.2 Dataset description and structure . . . 7
3.3 Anomaly categories . . . 7
3.4 Pixel-level class imbalance . . . 8
3.5 Initial observations . . . 10
3.6 Conclusion . . . 10

4 Data Preparation **11**
4.1 Introduction . . . 11
4.2 Mask conversion . . . 11
4.3 Image and mask resizing . . . 11
4.4 Training, validation and test split . . . 12
4.5 Data augmentation . . . 13
4.6 Labels for the complementary classification task . . . 13

4.7 Conclusion . . . 14

5 Modeling **15**
5.1 Introduction . . . 15
5.2 Modeling strategy . . . 15
5.3 Image classification models (DSO1) . . . 16
5.4 Anomaly segmentation models (DSO2) . . . 16
5.4.1 U-Net trained from scratch . . . 16
5.4.2 U-Net with a pretrained ResNet34 encoder . . . 16
5.4.3 DeepLabV3+ with a pretrained ResNet34 encoder . . . 17
5.4.4 Convolutional autoencoder . . . 17
5.4.5 Loss function and training configuration . . . 18
5.5 Conclusion . . . 18

6 Evaluation **19**
6.1 Introduction . . . 19
6.2 Evaluation protocol . . . 19
6.2.1 Single-split evaluation . . . 20
6.2.2 Five-fold grouped cross-validation . . . 20
6.2.3 Evaluation scope . . . 20
6.3 Experimental Results . . . 20
6.3.1 Validation results . . . 21
6.3.2 Held-out test results . . . 21
6.3.3 Five-fold cross-validation results . . . 22
6.3.4 Anomaly-type classification results . . . 23
6.4 Analysis and Discussion . . . 24
6.4.1 Qualitative analysis . . . 24
6.4.2 Failure cases . . . 25
6.4.3 Strengths of the proposed experiments . . . 26
6.4.4 Weaknesses and interpretation . . . 26
6.5 Limitations and Future Improvements . . . 27
6.5.1 Limitations of this study . . . 27
6.5.2 Possible improvements . . . 27
6.6 Conclusion . . . 28

7 Deployment **29**
7.1 Introduction . . . 29
7.2 Choice of the deployed model . . . 29
7.3 Description of the application . . . 29
7.4 Demonstration . . . 30

7.5 Potential practical use 33
7.6 Limits and improvements 33
7.7 Conclusion 33

General Conclusion **34**

References **35**
List of Abbreviations 35

List of Figures

3.1 Examples of original images, anomaly masks and image-mask overlays. . . 9
3.2 Number of images per anomaly category and distribution of anomaly area as a percentage of image pixels. Most anomalies occupy less than 10% of an image. 9

4.1 Examples of augmented training images and their corresponding masks. Geometric transformations are applied to both, while color transformations are applied only to the images. 13

6.1 Training and validation loss, validation Dice and validation IoU for the scratch U-Net. 21
6.2 Comparison of test Dice scores and per-fold Dice scores for the pretrained segmentation models. 22
6.3 Confusion matrices of the two classifiers on the held-out test split. 24
6.4 Test predictions of U-Net+ResNet34. Green indicates the ground-truth anomaly region and red indicates the predicted region. 25

7.1 Screenshots of the prototype on an animal image and a cone image. 31
7.2 Screenshots of the prototype on an obstacle image and a construction-scene image. 32

List of Tables

2.1 Business objectives and corresponding Data Science objectives. 5

3.1 Number of images per anomaly category. 8

4.1 Category distribution and mean anomaly area of the split (seed 183). . . . 12

5.1 Training configuration of the segmentation models. 18

6.1 Results on the validation set (14 images). 21
6.2 Results on the test set (12 images, segmentation threshold 0.5). 22
6.3 Five-fold grouped cross-validation results (mean ± standard deviation). . . 22
6.4 Test Dice score for each cross-validation fold. 23
6.5 Anomaly-type classification results across five folds. 23
6.6 Per-image Dice scores for selected test images, sorted by the U-Net+ResNet34 score. 25

General Introduction

Autonomous-driving systems rely on perception models that recognize a predefined set of semantic classes such as roads, cars, pedestrians and traffic signs. Real roads, however, contain unexpected objects that are absent from the training data: animals, rocks, traffic cones, lost tires and other obstacles. Failing to detect such objects is a serious safety risk, which makes road anomaly detection an important problem for intelligent vehicles.

The global objective of this project is to contribute to the road safety of autonomous vehicles by detecting these unexpected objects. It is broken down into two Data Science objectives: recognizing the type of anomaly present in a road scene through image classification, and precisely locating the anomalous regions through pixel-level semantic segmentation. The study is carried out on the RoadAnomaly dataset, which contains only 60 annotated images. The small size of the dataset makes the project a useful testbed for the deep learning techniques studied in the module, in particular convolutional neural networks, transfer learning and autoencoders.

The work follows the CRISP-DM methodology. For the classification objective, a small CNN trained from scratch is compared with a ResNet34 pretrained on ImageNet. For the segmentation objective, four approaches are compared: a U-Net trained from scratch, a U-Net with an ImageNet-pretrained ResNet34 encoder, DeepLabV3+ with the same encoder, and a convolutional autoencoder based on reconstruction error. To limit data leakage between similar scenes, the evaluation uses a scene-grouped split and 5-fold grouped cross-validation, with precision, recall, F1-score/Dice and IoU as the main metrics.

The report is organized as follows. Chapter 1 introduces the project and the CRISP-DM methodology. Chapter 2 defines the problem, the global objective and the business and Data Science objectives. Chapter 3 describes the dataset, and Chapter 4 explains how the data are prepared. Chapter 5 presents the classification and segmentation models and their training configuration. Chapter 6 describes the evaluation protocol, reports the experimental results, analyzes strengths, weaknesses and failure cases, and presents the limitations and future improvements. Chapter 7 discusses deployment considerations. Finally, the general conclusion summarizes the main findings.

Chapter 1: Project Overview and Methodology

1.1 Introduction

This chapter presents the methodological framework of the project. Before describing the data and the models, it is necessary to explain how the work is organized and why a structured process is used.

1.2 The CRISP-DM framework

This report follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology. CRISP-DM provides a structured framework for carrying out a data-driven project, from understanding the problem to evaluating the developed solution and considering its deployment.

The project is organized into the following six phases:

1. **Business Understanding:** define the road-safety problem, the project objectives and the expected outcomes.
2. **Data Understanding:** inspect the RoadAnomaly dataset, its image structure, semantic masks, anomaly categories and class imbalance.
3. **Data Preparation:** convert masks, resize and normalize images, create data splits and apply data augmentation.
4. **Modeling:** implement and train CNN-based segmentation models, transfer-learning approaches, an autoencoder and complementary classifiers.
5. **Evaluation:** assess the models using quantitative metrics, cross-validation and visual inspection of predictions.
6. **Deployment:** describe how the selected model could be integrated into a practical road-scene monitoring system and identify the additional work required before real-world use.

The phases are presented in separate sections for clarity. In practice, CRISP-DM is iterative: evaluation results can lead to new data-preparation choices, model adjustments or additional experiments.

1.3 Conclusion

CRISP-DM gives the report a clear and reproducible structure. The following chapters follow its six phases in order, starting with the business understanding of the road-safety problem.

Chapter 2: Business Understanding

2.1 Introduction

This chapter corresponds to the first CRISP-DM phase. It defines the road-safety problem, states the global objective, translates the business objectives into Data Science objectives and describes the expected outcomes of the project.

2.2 Problem definition

Autonomous-driving systems commonly rely on models trained to recognize a predefined set of semantic classes, such as roads, cars, pedestrians and traffic signs. Real road environments, however, may contain unexpected objects that are not represented in the training data. Examples include animals, rocks, traffic cones, lost tires and other obstacles.

A conventional semantic segmentation model may incorrectly assign an unfamiliar object to one of its known classes. In a road-safety context, failing to identify such an object can create a risk for the vehicle and its passengers. Road anomaly detection therefore aims to identify and localize regions that differ from the expected road-scene content.

In this project, the problem is addressed at two levels: an image-level classification of the road scene, and a pixel-level **binary semantic segmentation task** in which each pixel of the image is classified as either anomalous or non-anomalous.

2.3 Global objective

The global objective of the project is to **ensure the road safety of autonomous vehicles by detecting unexpected objects** (animals, rocks, cones, debris) present on the roadway.

2.4 Business objectives and Data Science objectives

To reach this global objective, two business objectives (BO) are defined. Each of them is translated into a Data Science objective (DSO) that specifies the model to build and the

type of task to solve, as summarized in Table 2.1.

Business objective (BO)	**Data Science objective (DSO)**	**Type of task**
BO1. Quickly identify the type of unexpected obstacle present in a road scene, in order to trigger an appropriate alert.	**DSO1.** Build a classification model (CNN with transfer learning) that identifies the type of anomaly present in an image (animals, obstacles or other).	Image classification
BO2. Precisely locate the obstacle on the road, so that the vehicle can brake or avoid it.	**DSO2.** Build a semantic segmentation model that produces a pixel-wise mask of the anomalous regions.	Segmentation

Table 2.1: Business objectives and corresponding Data Science objectives.

BO1 concerns the *recognition* of a hazard. It requires a fast, image-level decision about the kind of obstacle present, so that an appropriate alert can be triggered. This corresponds to an image classification task (**DSO1**). Since every image of the RoadAnomaly dataset contains an anomaly, the classes are the anomaly categories: animals, obstacles, and other (cones, lost tires and vehicles).

BO2 concerns the *localization* of the hazard. Knowing that an obstacle exists is not sufficient for the vehicle to react: its position and extent on the road must also be known. This corresponds to a semantic segmentation task (**DSO2**), whose output is a binary mask of the anomalous regions. Segmentation is the central task of this project.

2.5 Project objectives

The specific objectives are:

- Explore the RoadAnomaly dataset and understand the relationship between each image and its ground-truth mask.
- Prepare the images and binary masks for deep learning experiments.
- Build training, validation and test subsets while limiting data leakage between related scenes.
- Implement a CNN-based classification model using transfer learning to identify the anomaly category (DSO1).
- Implement and compare CNN-based segmentation approaches, including a model trained from scratch and transfer-learning models (DSO2).

- Explore a convolutional autoencoder as a complementary anomaly-detection approach.
- Analyze prediction visualizations, failure cases, limitations and possible improvements.

2.6 Expected outcomes and success criteria

The expected outcome is a documented experimental pipeline that takes a road image as input, predicts the category of the anomaly and produces a pixel-level anomaly prediction. The project also aims to determine which of the investigated approaches is more suitable for the available dataset.

The main success criteria are:

- correct correspondence between original images and their ground-truth masks;
- a reproducible data-preparation and evaluation protocol;
- meaningful localization of the anomalous regions on the road;
- a comparison of models trained from scratch, pretrained models and the autoencoder;
- an honest analysis of generalization errors, especially on scenes not represented in training.

Because the dataset is small, the objective is not to claim that the developed system is ready for deployment in a real autonomous vehicle. The objective is to build and evaluate a proof of concept and identify the conditions required for a more reliable system.

2.7 Conclusion

The global objective of road safety is broken down into two business objectives, recognizing and locating unexpected obstacles, which are translated into an image classification task (DSO1) and a semantic segmentation task (DSO2). These objectives guide the data exploration presented in the next chapter.

Chapter 3: Data Understanding

3.1 Introduction

This chapter corresponds to the second CRISP-DM phase. It describes the RoadAnomaly dataset, its annotations and its main statistical properties, which directly influence the choices made in the following chapters.

3.2 Dataset description and structure

The RoadAnomaly dataset used in this project contains **60 images** in WebP format. All images have a resolution of 1280×720 pixels. Each image has a corresponding semantic label file named `labels_semantic.png`. We verified that all 60 images have a matching mask.

The masks contain two values, $\{0, 2\}$. The value 2 identifies anomalous pixels, while 0 represents all remaining pixels. We convert these annotations into binary masks for the segmentation task:

$$y(x, y) = \begin{cases} 1 & \text{if the pixel is anomalous,} \\ 0 & \text{otherwise.} \end{cases}$$

Consequently, the main task is binary pixel-level segmentation rather than multi-class semantic segmentation.

3.3 Anomaly categories

The anomaly category of each image is identified from its file name. Table 3.1 presents the distribution of the 60 images.

Category	Number of images
Animals	31
Obstacles (rocks, Czech hedgehog, etc.)	15
Cones	8
Lost tires	3
Vehicle	3
Total	**60**

Table 3.1: Number of images per anomaly category.

The dataset is dominated by animal examples. The lost-tire and vehicle categories contain only three images each. This imbalance makes it difficult to train a reliable classifier for every individual category.

3.4 Pixel-level class imbalance

The dataset also presents a strong pixel-level imbalance. Anomalous regions cover, on average, only **9.85%** of the pixels in an image. The remaining pixels correspond to non-anomalous content.

The size and shape of anomalies vary considerably. Some images contain large animals close to the camera, while others contain small rocks or obstacles that occupy only a limited area. This imbalance is important because a model could obtain a seemingly good pixel accuracy by predicting mostly background. For this reason, the experiments focus on overlap-based metrics such as Dice and IoU, in addition to precision and recall.

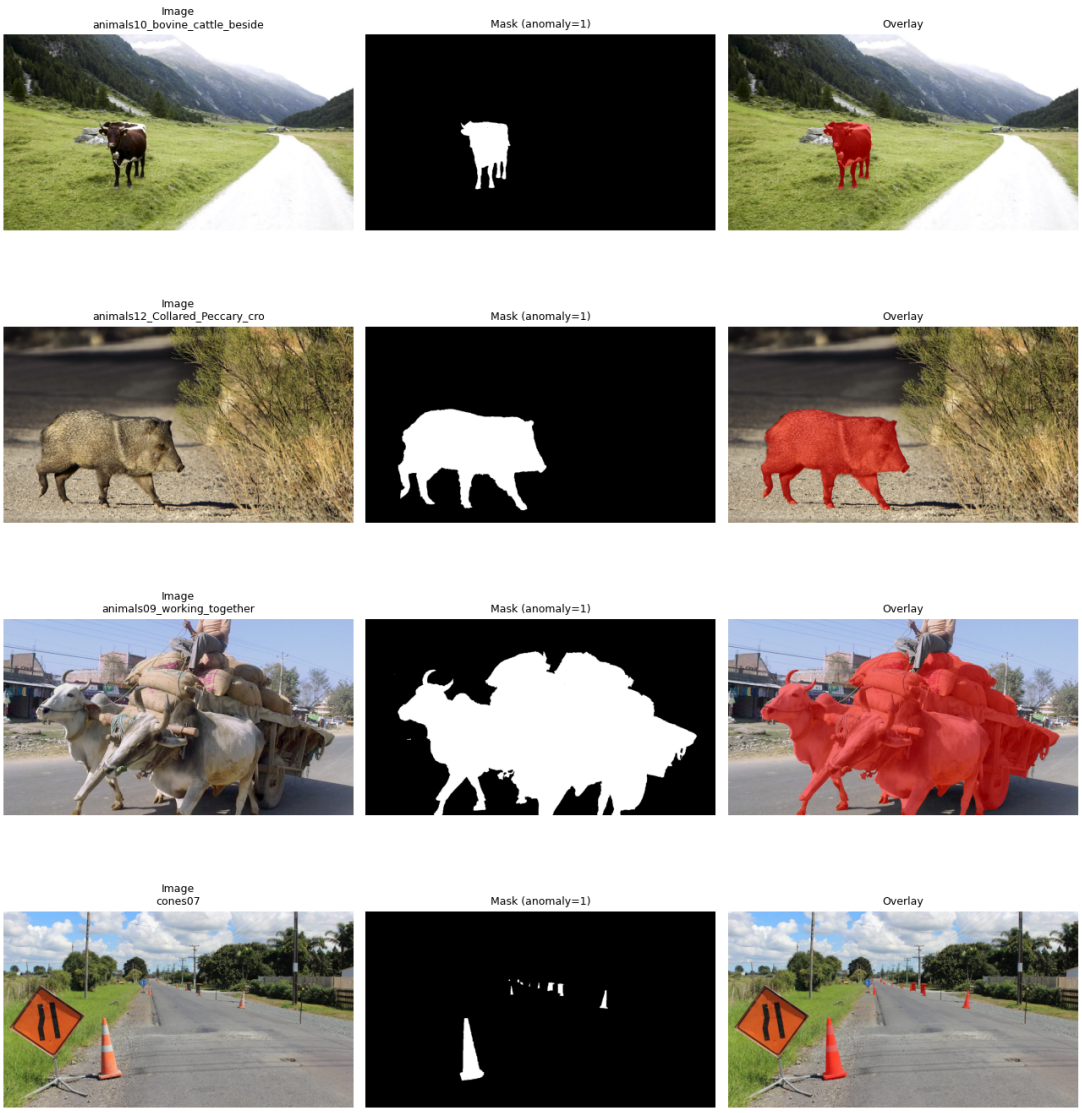


Figure 3.1: Examples of original images, anomaly masks and image-mask overlays.

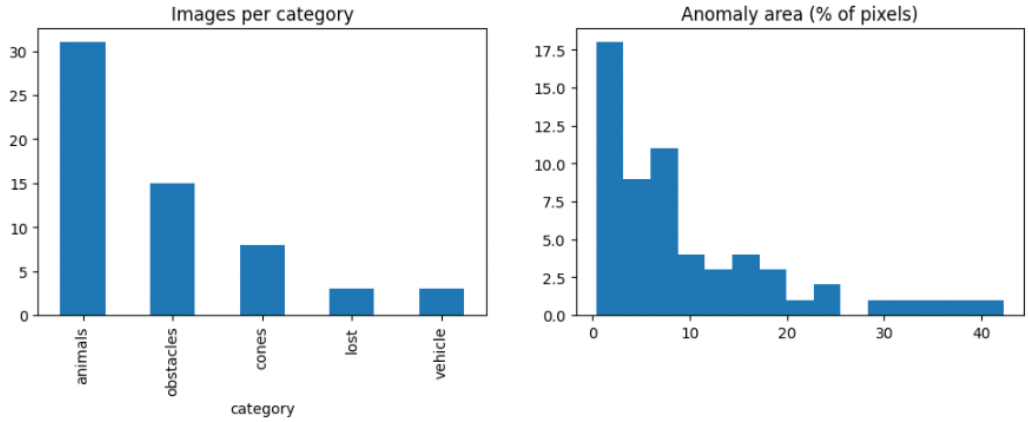


Figure 3.2: Number of images per anomaly category and distribution of anomaly area as a percentage of image pixels. Most anomalies occupy less than 10% of an image.

3.5 Initial observations

The dataset is suitable for studying binary anomaly segmentation because every image has a pixel-level annotation. However, the limited number of images, the imbalance between categories and the variation in anomaly size make the task challenging.

These observations influence the next CRISP-DM phase. In particular, the data preparation process must preserve the exact image-mask correspondence, and the evaluation split must avoid placing highly similar images from the same scene in both training and test subsets.

3.6 Conclusion

The dataset is small, dominated by animal examples and strongly imbalanced at pixel level. These characteristics motivate overlap-based metrics, scene-grouped splitting and data augmentation, which are addressed in the data preparation phase.

Chapter 4: Data Preparation

4.1 Introduction

This chapter corresponds to the third CRISP-DM phase. It explains how the raw images and masks are transformed into inputs suitable for deep learning, how the data are split without leakage and how augmentation is used to compensate for the small training set.

4.2 Mask conversion

The original masks use the value 2 to identify anomalous pixels and 0 for the remaining pixels. Before training, each mask is converted into a binary representation: anomalous pixels become 1 and all other pixels become 0.

This conversion is necessary because the segmentation models use a single output channel and a binary prediction objective.

4.3 Image and mask resizing

The original images and masks are resized from 1280×720 to 512×288, preserving the original aspect ratio.

Different interpolation methods are used:

- **Bilinear interpolation** is applied to RGB images because it provides smooth image resizing.
- **Nearest-neighbor interpolation** is applied to masks to preserve discrete label values and avoid creating invalid intermediate labels.

After resizing, images are normalized using the ImageNet mean and standard deviation. This normalization is required by the pretrained encoders and is applied consistently across the segmentation models.

4.4 Training, validation and test split

Several images originate from the same scene. A purely random split could therefore place visually similar frames in both training and test sets, resulting in overly optimistic evaluation scores.

To reduce this risk, a grouped and stratified split is created using `StratifiedGroupKFold` with five folds:

- Images belonging to the same scene are assigned to the same group. The group is derived from the image file name.
- Stratification is based on the anomaly category. The small categories (cones, lost tires and vehicles) are merged into an "other" category for stratification only.
- One fold is used for testing, another for validation, and the three remaining folds are used for training.

Among 200 random seeds, the seed producing the most balanced split was selected. The selection criterion considers category presence and the mean anomaly area in each subset, not model performance. We also verify that no scene group appears in more than one subset.

The resulting split contains 34 training images, 14 validation images and 12 test images, as shown in Table 4.1.

Category	**Train**	**Validation**	**Test**
Animals	21	6	4
Cones	6	1	1
Lost tires	1	1	1
Obstacles	5	5	5
Vehicle	1	1	1
Total	**34**	**14**	**12**
Mean anomaly area	**9.6%**	**11.2%**	**8.8%**

Table 4.1: Category distribution and mean anomaly area of the split (seed 183).

An important consequence of grouping by scene is that some scene types, particularly rock fields, are absent from the training subset and appear in the test subset. This makes the test evaluation more challenging but provides a more realistic indication of generalization to different scenes.

4.5 Data augmentation

Because the training set contains only 34 images, data augmentation is applied to increase the diversity of training examples. Albumentations is used for the transformations [**?**].

The transformations include:

- horizontal flip with probability $p = 0.5$;
- affine transformation with scale between 0.8 and 1.2, translation up to 10%, rotation of $\pm 10°$ and probability $p = 0.7$;
- random brightness and contrast with probability $p = 0.5$;
- hue and saturation changes with probability $p = 0.3$.

Geometric transformations are applied identically to the image and its mask so that the pixel-level correspondence is preserved. Color transformations are applied only to the image.

The training indices are repeated four times. Consequently, each epoch contains 136 augmented samples, corresponding to four augmented versions of each of the 34 original training images.



Figure 4.1: Examples of augmented training images and their corresponding masks. Geometric transformations are applied to both, while color transformations are applied only to the images.

4.6 Labels for the complementary classification task

In addition to segmentation, an image-level classification task is considered. Each image is assigned to one of three categories:

- **Animals**;
- **Obstacles**;
- **Other**, which combines cones, lost tires and vehicles.

The categories with very few examples are grouped into “other” because there are not enough images to train separate classifiers for them. This class is therefore heterogeneous.

4.7 Conclusion

The data are now converted, resized, normalized, split by scene and augmented. This protocol preserves the image-mask correspondence and limits data leakage, so the models presented in the next chapter can be trained and compared fairly.

Chapter 5: Modeling

5.1 Introduction

This chapter corresponds to the fourth CRISP-DM phase. It presents the models built to answer the two Data Science objectives defined in Chapter 2: an image-level classification model (DSO1) and pixel-level segmentation models (DSO2). For each objective, the chapter describes the selected architectures and the training configuration.

5.2 Modeling strategy

The two Data Science objectives are addressed by two families of models:

- **DSO1 – image classification:** two classifiers predict the anomaly category of a whole image (animals, obstacles or other): a small CNN trained from scratch and a CNN with transfer learning.
- **DSO2 – anomaly segmentation:** four approaches produce a pixel-level anomaly mask. They were selected to investigate different deep learning strategies studied in the module:
  - a CNN segmentation model trained from scratch;
  - transfer learning with an ImageNet-pretrained encoder;
  - a segmentation architecture designed to capture information at multiple spatial scales;
  - an autoencoder that uses reconstruction error as an anomaly score.

In both cases, models trained from scratch serve as baselines for measuring the benefit of transfer learning. All models are implemented in PyTorch. The pretrained segmentation architectures use the `segmentation_models_pytorch` library.

5.3 Image classification models (DSO1)

The classification task predicts one label for an entire image rather than a mask for each pixel. The labels are the three categories defined during data preparation: animals, obstacles and other (cones, lost tires and vehicles). Two classifiers are compared:

- **Small CNN trained from scratch:** a network composed of four convolutional blocks followed by global average pooling. It serves as the baseline.
- **ResNet34 with transfer learning:** a ResNet34 pretrained on ImageNet [**?**], in which the original classifier is replaced by a new three-class classification head. The pretrained layers provide general visual features such as edges, textures and object parts, which is valuable when only a few dozen training images are available.

Class-weighted cross-entropy is used to compensate for the imbalance between the three classes. The model checkpoint is selected using the validation macro-F1 score. The classifiers are trained and evaluated on the same scene-grouped splits as the segmentation models.

5.4 Anomaly segmentation models (DSO2)

5.4.1 U-Net trained from scratch

U-Net is an encoder-decoder architecture originally introduced for biomedical image segmentation [**?**]. Its encoder progressively extracts higher-level features while reducing spatial resolution. Its decoder restores spatial resolution to produce a segmentation mask.

U-Net uses skip connections that transfer feature maps from the encoder to the decoder. These connections help recover spatial details that may be lost during downsampling, which is useful when localizing small objects.

The baseline implemented in this project has five levels with 16, 32, 64, 128 and 256 channels. It uses batch normalization and transposed convolutions for upsampling. The model contains approximately 1.94 million parameters.

This model provides a baseline for measuring the benefit of transfer learning.

5.4.2 U-Net with a pretrained ResNet34 encoder

Training a deep feature extractor from scratch is difficult when only 34 training images are available. To address this limitation, the U-Net encoder is replaced with a ResNet34 pretrained on ImageNet .

The pretrained encoder provides general visual features, such as edges, textures and object parts. The decoder is trained to combine these features and produce the binary

anomaly mask.

The model contains approximately 24.4 million parameters. Differential learning rates are used:

- 10^{-4} for the pretrained encoder, to limit large changes to its learned features;
- 10^{-3} for the randomly initialized decoder, which must learn the segmentation task.

This approach is selected to evaluate whether pretrained visual representations improve segmentation when the target dataset is very small.

5.4.3 DeepLabV3+ with a pretrained ResNet34 encoder

DeepLabV3+ is an encoder-decoder architecture for semantic segmentation. It uses atrous (dilated) convolutions and an Atrous Spatial Pyramid Pooling (ASPP) module to capture contextual information at multiple scales.

This multi-scale representation is relevant to road anomaly detection because anomalies vary considerably in size, from small rocks to large animals.

DeepLabV3+ uses the same pretrained ResNet34 encoder and a comparable training procedure to the previous model. This makes it possible to compare the effect of the decoder architecture while keeping the feature extractor and main training choices similar.

5.4.4 Convolutional autoencoder

The convolutional autoencoder explores a different anomaly-detection strategy. Instead of directly learning to segment known anomaly examples, it is trained to reconstruct normal road-scene content. The reconstruction error is then used as an anomaly score: regions that are reconstructed poorly may correspond to unexpected objects.

The implemented architecture contains four stride-2 convolutional layers, which progressively downsample the input by a factor of 16. A 1×1 bottleneck with 16 channels is followed by a symmetric decoder.

To reduce the influence of anomalous pixels during training, those pixels are excluded from the reconstruction loss using a masked mean squared error (MSE). Since ground-truth anomaly masks are used to exclude these pixels, this implementation is **semi-supervised rather than fully unsupervised**.

At inference time, the squared reconstruction error is averaged across the color channels, smoothed using a 15×15 average filter and thresholded to obtain a binary anomaly mask. The threshold is selected on the validation set to maximize Dice and is then kept fixed for the test evaluation.

5.4.5 Loss function and training configuration

For the segmentation models, the training objective combines binary cross-entropy (BCE) and soft Dice loss:

$$\mathcal{L} = \mathrm{BCE}(p, y) + \left(1 - \frac{2\sum p\, y + 1}{\sum p + \sum y + 1}\right), \tag{5.1}$$

where p represents the predicted probabilities and y represents the ground-truth binary mask.

BCE provides a pixel-wise learning signal, while the Dice component encourages overlap between predicted and ground-truth anomaly regions. The Dice component is particularly useful when the foreground class represents only a small proportion of the image.

The segmentation models use AdamW with a weight decay of 10^{-4} and a cosine learning-rate schedule. The checkpoint with the best validation Dice is retained.

Configuration	**U-Net scratch**	**U-Net+R34**	**DeepLabV3+**	**Autoencoder**
Parameters	1.94M	24.4M	–	–
Pretraining	None	ImageNet	ImageNet	None
Learning rate	10^{-3}	$10^{-4}/10^{-3}$	$10^{-4}/10^{-3}$	10^{-3} (Adam)
Epochs	60	40	40	40
Batch size	4	4	4	4
Loss	BCE+Dice	BCE+Dice	BCE+Dice	Masked MSE

Table 5.1: Training configuration of the segmentation models.

5.5 Conclusion

Two classifiers (DSO1) and four segmentation approaches (DSO2) have been defined. Each objective includes a model trained from scratch as a baseline and a transfer-learning model, and the segmentation part additionally covers multi-scale segmentation and reconstruction-based scoring. The next chapter describes the protocol used to evaluate them.

Chapter 6: Evaluation

6.1 Introduction

This chapter corresponds to the fifth CRISP-DM phase. It defines the metrics and evaluation protocols, reports the experimental results, analyzes the predictions and failure cases, and discusses the limitations of the study together with possible improvements.

6.2 Evaluation protocol

For the segmentation models, predicted probabilities are converted into binary masks using a threshold of 0.5. The predictions are compared with the ground-truth masks at pixel level.

Let TP, FP and FN represent the numbers of true-positive, false-positive and false-negative pixels. The evaluation metrics are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{6.1}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{6.2}$$

$$\text{F1-score} = \frac{2TP}{2TP + FP + FN}, \tag{6.3}$$

$$\text{Dice} = \frac{2TP}{2TP + FP + FN}, \tag{6.4}$$

$$\text{IoU} = \frac{TP}{TP + FP + FN}. \tag{6.5}$$

At pixel level, F1-score and Dice are mathematically equivalent, so they are reported as one metric.

- **Precision** measures the proportion of predicted anomaly pixels that are actually anomalous.
- **Recall** measures the proportion of ground-truth anomaly pixels that the model successfully detects.

- **F1-score/Dice** balances precision and recall and measures the overlap between the prediction and the ground truth.
- **IoU** measures the intersection divided by the union of the predicted and ground-truth anomaly regions.
- **AUROC** evaluates how well an anomaly score separates anomalous and non-anomalous pixels across different thresholds. It is additionally reported for the autoencoder.

We also compute mean per-image IoU. For the autoencoder, pixel-level AUROC is reported in addition to threshold-dependent segmentation metrics.

6.2.1 Single-split evaluation

The first evaluation uses the grouped split described in Chapter 4: 34 training images, 14 validation images and 12 test images.

The validation set is used for checkpoint selection and for selecting the autoencoder threshold. The test set is reserved for the final evaluation and is not used to select model parameters.

6.2.2 Five-fold grouped cross-validation

A single test split of only 12 images can produce a noisy estimate of model performance. To obtain a broader view, 5-fold grouped and stratified cross-validation is also conducted.

Each fold is used once as the test fold, while the next fold is used for validation and the remaining folds are used for training. This ensures that every image is evaluated once as part of a test fold while keeping images from the same scene together.

The mean and standard deviation of the metrics across the five folds are reported. This provides information about both the average performance and its variability.

6.2.3 Evaluation scope

Dice and IoU are the principal segmentation metrics in this project, as required by the module assessment. The standard RoadAnomaly benchmark metrics, such as pixel-level AuPRC and FPR95, are not included. Consequently, the reported results should not be interpreted as directly comparable with published benchmark scores.

6.3 Experimental Results

This section reports the quantitative results obtained with the protocol described above: validation results, held-out test results, five-fold cross-validation and anomaly-type classifi-

cation.

6.3.1 Validation results

Table 6.1 presents the results on the 14-image validation set. Transfer learning substantially improves the validation Dice compared with the scratch U-Net.

Model	Precision	Recall	F1 / Dice	IoU
U-Net scratch	0.554	0.505	0.528	0.359
U-Net + ResNet34	0.905	0.855	0.880	0.785
DeepLabV3+ ResNet34	0.959	0.825	0.887	0.797
Autoencoder	0.131	0.581	0.214	0.120

Table 6.1: Results on the validation set (14 images).

These validation values should be interpreted carefully because the best epoch is selected using a small set of only 14 images. For example, the validation Dice of U-Net+ResNet34 fluctuates between approximately 0.75 and 0.88 across epochs. The held-out test results and cross-validation provide additional information about generalization.

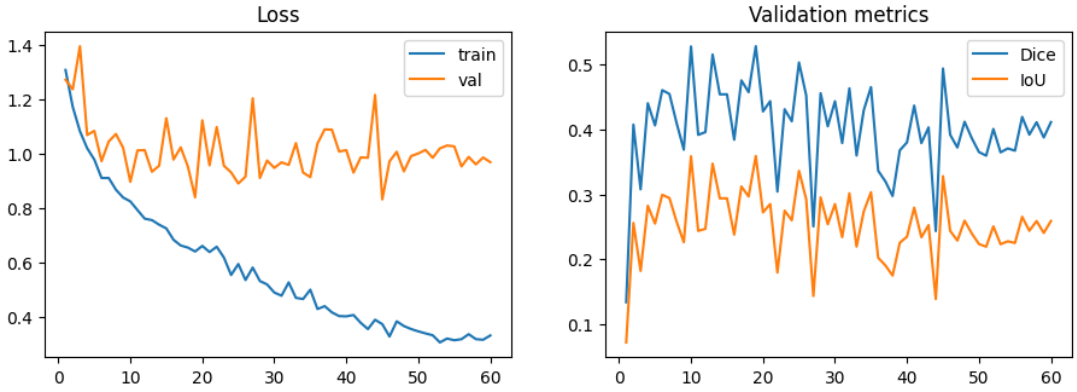


Figure 6.1: Training and validation loss, validation Dice and validation IoU for the scratch U-Net.

6.3.2 Held-out test results

Table 6.2 presents the results on the 12 test images. The model weights are selected using validation performance, and the autoencoder threshold is fixed using the validation set.

Model	Precision	Recall	F1 / Dice	IoU
U-Net scratch	0.671	0.524	0.589	0.417
U-Net + ResNet34	0.708	0.623	0.663	0.496
DeepLabV3+ ResNet34	**0.930**	0.648	**0.764**	**0.618**
Autoencoder	0.121	**0.698**	0.207	0.115

Table 6.2: Results on the test set (12 images, segmentation threshold 0.5).

DeepLabV3+ obtains the highest test precision, Dice and IoU on this particular split. Its recall is 0.648, which indicates that some anomalous pixels remain undetected. The autoencoder has a higher recall than the other models but very low precision, meaning that it predicts many non-anomalous pixels as anomalies.

The test results are lower than the validation results. One explanation is that the test set contains scene types, including rock fields and the Czech hedgehog scene, that are not represented in the training subset. The autoencoder pixel AUROC is 0.564 on validation and 0.660 on test, which indicates limited separation between anomalous and non-anomalous pixels.

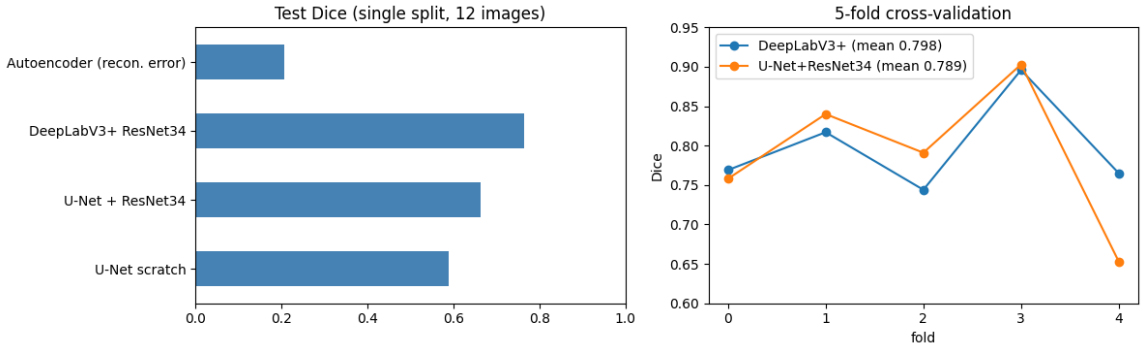


Figure 6.2: Comparison of test Dice scores and per-fold Dice scores for the pretrained segmentation models.

6.3.3 Five-fold cross-validation results

Table 6.3 presents the mean and standard deviation of the two pretrained segmentation models across five grouped folds.

Model	Precision	Recall	Dice	IoU
U-Net + ResNet34	0.839 ± 0.157	0.764 ± 0.103	0.789 ± 0.094	0.659 ± 0.126
DeepLabV3+	0.874 ± 0.082	0.745 ± 0.105	0.798 ± 0.061	0.668 ± 0.088

Table 6.3: Five-fold grouped cross-validation results (mean ± standard deviation).

The average Dice scores are close: 0.789 for U-Net+ResNet34 and 0.798 for DeepLabV3+. The difference is small compared with the variation across folds. Therefore, the experiments do not provide strong evidence that one of these two architectures is consistently better than the other.

Model	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
U-Net + ResNet34	0.758	0.840	0.791	0.903	0.653
DeepLabV3+	0.769	0.817	0.744	0.896	0.765

Table 6.4: Test Dice score for each cross-validation fold.

U-Net+ResNet34 performs better in three folds out of five, while DeepLabV3+ performs better in the other two. On the single split, DeepLabV3+ appears clearly better (0.764 against 0.663), but the cross-validation results show that this difference is not consistent across folds. A single evaluation on 12 images is therefore not sufficient to establish a definitive ranking.

6.3.4 Anomaly-type classification results

The complementary classification task predicts one of three image-level categories: animals, obstacles or other.

On the single test split, both classifiers obtain 58.3% accuracy. Their macro-F1 scores are 0.552 for the small CNN and 0.556 for ResNet34. However, the test split contains obstacle scenes, such as rock fields and the Czech hedgehog, that are absent from training. This affects obstacle recognition.

The cross-validation results in Table 6.5 provide a broader comparison.

	Majority baseline		Small CNN		ResNet34	
Fold	Acc.	–	Acc.	Macro-F1	Acc.	Macro-F1
0	0.333	–	0.417	0.317	0.583	0.544
1	0.429	–	0.357	0.354	0.857	0.878
2	0.800	–	0.500	0.427	0.800	0.786
3	0.545	–	0.364	0.277	0.909	0.604
4	0.538	–	0.538	0.528	0.769	0.692
Mean ± std	0.529 ± 0.175	–	0.435 ± 0.081	0.381 ± 0.099	0.784 ± 0.124	0.701 ± 0.135

Table 6.5: Anomaly-type classification results across five folds.

The pretrained ResNet34 obtains a higher macro-F1 than the small CNN in all five folds. The small CNN remains below the majority-class baseline in average accuracy, suggesting that it does not learn sufficiently useful category representations from this

limited training data. The fold sizes are small, so the results should be interpreted as an observed trend rather than a precise estimate of classification performance.

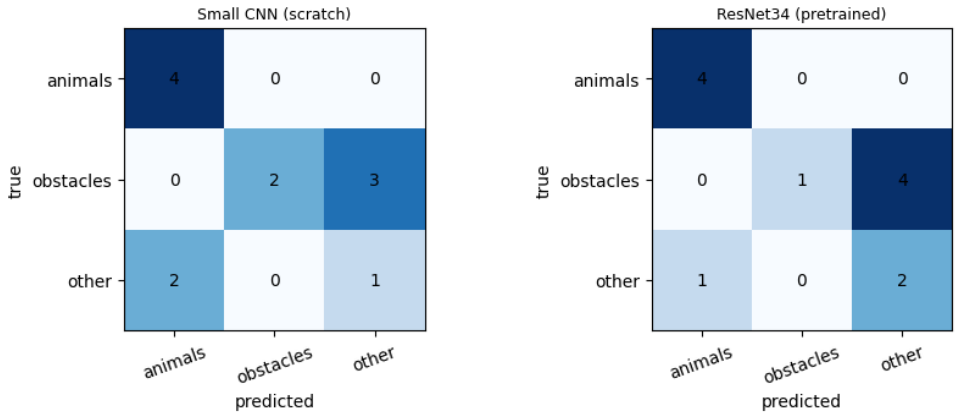


Figure 6.3: Confusion matrices of the two classifiers on the held-out test split.

Overall, pretrained models clearly outperform the scratch U-Net and the autoencoder. DeepLabV3+ leads on the single test split, but the cross-validation shows that the two pretrained architectures are statistically close.

6.4 Analysis and Discussion

Numerical scores summarize performance but do not explain it. This section examines the predictions visually, studies the failure cases and discusses the strengths and weaknesses of the experiments.

6.4.1 Qualitative analysis

Quantitative metrics provide an overall measure of segmentation quality, but visual inspection is also necessary to understand where the models succeed and fail.

Figure 6.4 presents predictions from U-Net+ResNet34 on the 12 test images. Large and compact objects, such as animals, cones and vehicles, are generally localized more successfully. In several examples, the predicted boundaries are close to the ground-truth annotations. However, performance varies significantly between scenes.

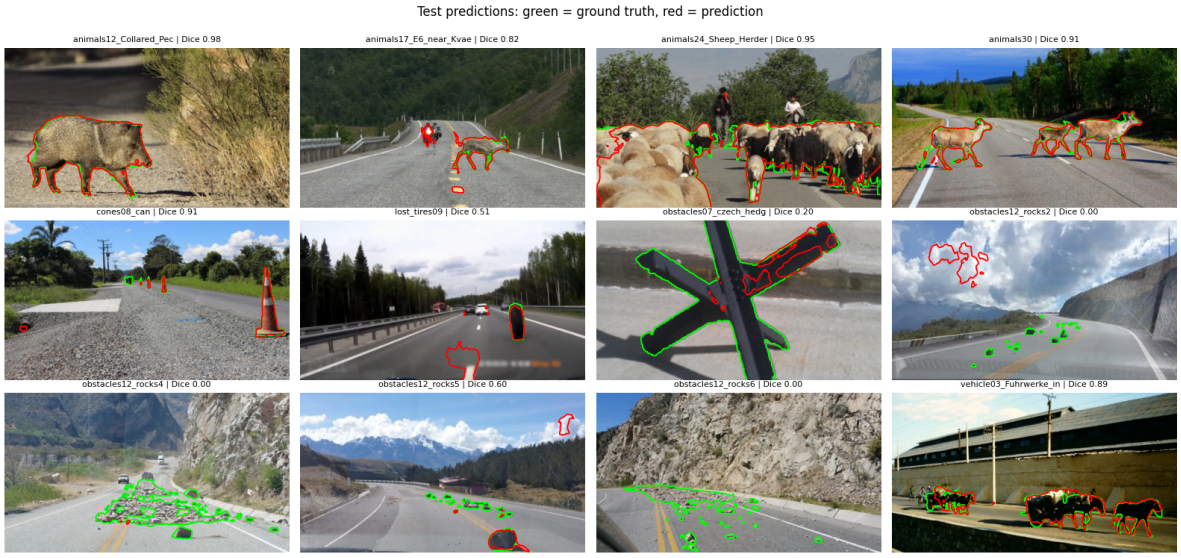


Figure 6.4: Test predictions of U-Net+ResNet34. Green indicates the ground-truth anomaly region and red indicates the predicted region.

6.4.2 Failure cases

Table 6.6 presents selected test images with low per-image Dice scores.

Image	Category	U-Net scratch	U-Net+R34	DeepLabV3+
`obstacles12_rocks2`	Obstacles	0.24	0.00	0.05
`obstacles12_rocks6`	Obstacles	0.07	0.00	0.03
`obstacles07_czech_hedgehog`	Obstacles	0.72	0.07	0.22
`obstacles12_rocks4`	Obstacles	0.04	0.09	0.02
`animals17_E6_near_Kvaenangen`	Animals	0.52	0.58	0.76
`obstacles12_rocks5`	Obstacles	0.54	0.65	0.62
`cones08_can`	Cones	0.36	0.82	0.85
`animals30`	Animals	0.75	0.82	0.88

Table 6.6: Per-image Dice scores for selected test images, sorted by the U-Net+ResNet34 score.

Three important failure patterns can be identified:

- **Scattered small rocks:** for the images `rocks2`, `rocks4` and `rocks6`, Dice is at most 0.09 for the pretrained models. These rocks are small, numerous and visually similar to road texture. Resizing to 512×288 also removes some of their details. Since the rock-field scenes are absent from training, the models have difficulty generalizing to them.
- **Unusual large shape:** for the Czech hedgehog image, the pretrained models obtain Dice scores of 0.07 and 0.22, while the scratch U-Net obtains 0.72. This is a single example and does not establish a general advantage for training from scratch, but it demonstrates that pretraining does not improve every individual case.

- **Scene variation:** the difference between training and test scenes also contributes to classification errors, particularly for the obstacle category.

6.4.3 Strengths of the proposed experiments

- **Transfer learning improves segmentation performance.** The pretrained models achieve higher validation and cross-validation Dice scores than the scratch baseline.
- **The pretrained models achieve relatively high precision.** Their cross-validated precision ranges from approximately 0.84 to 0.87, indicating that many predicted anomaly pixels are correct.
- **The evaluation considers scene grouping.** Keeping related images in the same subset reduces the risk of overly optimistic results caused by similar frames appearing in both training and testing.
- **Cross-validation exposes performance variability.** Reporting standard deviations gives a more informative picture than relying only on one test split.
- **Several modeling strategies are compared.** The experiments include a baseline CNN, transfer learning, multi-scale segmentation and reconstruction-based anomaly scoring.

6.4.4 Weaknesses and interpretation

- **Recall is lower than precision for the pretrained segmentation models.** Their cross-validated recall is approximately 0.75, compared with precision between 0.84 and 0.87. The models therefore miss some anomalous pixels. In a road-safety application, missed obstacles are particularly important.
- **Performance depends on the scene.** Dice scores vary across folds, showing that generalization is sensitive to the visual content of the test subset.
- **The autoencoder is not competitive in this experiment.** It obtains a test Dice of 0.207, test AUROC of 0.660 and precision of 0.121. One possible explanation is that reconstruction error responds to visual complexity, such as vegetation, shadows or road markings, rather than only to unexpected objects. This explanation is a hypothesis and was not verified through a dedicated experiment.
- **The supervised models may have limited open-set generalization.** They are trained using the anomaly examples available in the dataset, which is dominated by animals. Their behavior on entirely new anomaly types remains uncertain.

In summary, the models localize large and compact anomalies well but struggle with small scattered rocks and scenes absent from training. Recall and scene dependence are the main weaknesses.

6.5 Limitations and Future Improvements

Every experimental study has boundaries. This section states the limitations of the present work and proposes directions that could improve both the results and the reliability of the evaluation.

6.5.1 Limitations of this study

The experiments have several limitations:

- **Small dataset:** the dataset contains only 60 images, and the held-out test set contains 12 images. A single image can therefore have a noticeable effect on the reported scores.
- **Small validation set:** selecting the best epoch using only 14 images can produce noisy validation estimates.
- **Scene distribution:** grouped splitting avoids leakage but can create difficult distribution differences between training and test subsets.
- **Autoencoder supervision:** the anomaly masks are used to exclude anomalous pixels from the reconstruction loss. The autoencoder is therefore semi-supervised, not fully unsupervised.
- **Metric coverage:** Dice and IoU are reported as required, but AuPRC and FPR95 are not included. The results are consequently not directly comparable with the complete published benchmark protocol.
- **Limited deployment validation:** the project evaluates models on the available dataset and does not include real-time testing on a vehicle or embedded platform.

The held-out test set is used for final evaluation and qualitative error analysis, but not for training or model-selection decisions. It also corresponds to one of the grouped folds in the cross-validation experiment; these evaluations should therefore not be interpreted as fully independent datasets.

6.5.2 Possible improvements

Several directions could improve the system:

- **Higher image resolution:** using a resolution such as 768×432, or applying image tiling, may preserve details of small objects such as rocks.
- **Recall-oriented losses:** Focal loss or Tversky loss could be investigated to reduce missed anomalous pixels.
- **More training data:** additional real road images and synthetic anomalies could improve the diversity of the training set.
- **Test-time augmentation and ensembling:** combining predictions from multiple transformations or models may improve robustness, although this must be measured experimentally.
- **Open-set anomaly detection:** uncertainty-based methods, feature-density approaches or image-resynthesis methods could help identify objects that do not belong to known training categories [**?**].
- **Repeated experiments:** training with several random seeds and reporting mean and standard deviation would provide more stable estimates.
- **Deployment optimization:** model compression, hardware-aware inference and latency measurement would be necessary for a real-time application.

6.6 Conclusion

The evaluation combines overlap-based metrics, a held-out test split and grouped cross-validation. Pretrained segmentation models clearly outperform the scratch U-Net and the autoencoder, with U-Net+ResNet34 and DeepLabV3+ close to each other in cross-validation. The main weaknesses are low recall, strong dependence on the scene and difficulty with small scattered rocks. The very small dataset and the incomplete metric coverage are the principal limitations, and higher resolution, recall-oriented losses, more data and open-set methods define the roadmap for future work.

Chapter 7: Deployment

7.1 Introduction

The last phase of CRISP-DM is the deployment of the solution. To show how the trained model could be used in practice, we built a prototype web application. This chapter justifies the model that was deployed, describes the application, illustrates it on example images and discusses its practical use and limits.

7.2 Choice of the deployed model

Two models reached similar performance in the evaluation: U-Net with a ResNet34 encoder and DeepLabV3+ with the same encoder (cross-validated Dice 0.789 ± 0.094 and 0.798 ± 0.061). The difference is far smaller than the variation across folds, so neither is significantly better. We deployed DeepLabV3+ for three reasons:

- it obtained the best Dice on the test split (0.764) and the highest cross-validated Dice and precision;
- its results are more stable across folds (standard deviation of the Dice 0.061 against 0.094), which suggests it is less sensitive to the choice of training images;
- its ASPP module captures context at several scales, which suits anomalies whose size varies from small cones to large animals.

This choice has a cost: its recall (0.745) is slightly lower than that of U-Net+ResNet34 (0.764), so it misses somewhat more of the anomalous pixels. For a safety application this trade-off would need to be reconsidered. The deployed weights are [state exactly which checkpoint: run, fold, or retraining on training+validation].

7.3 Description of the application

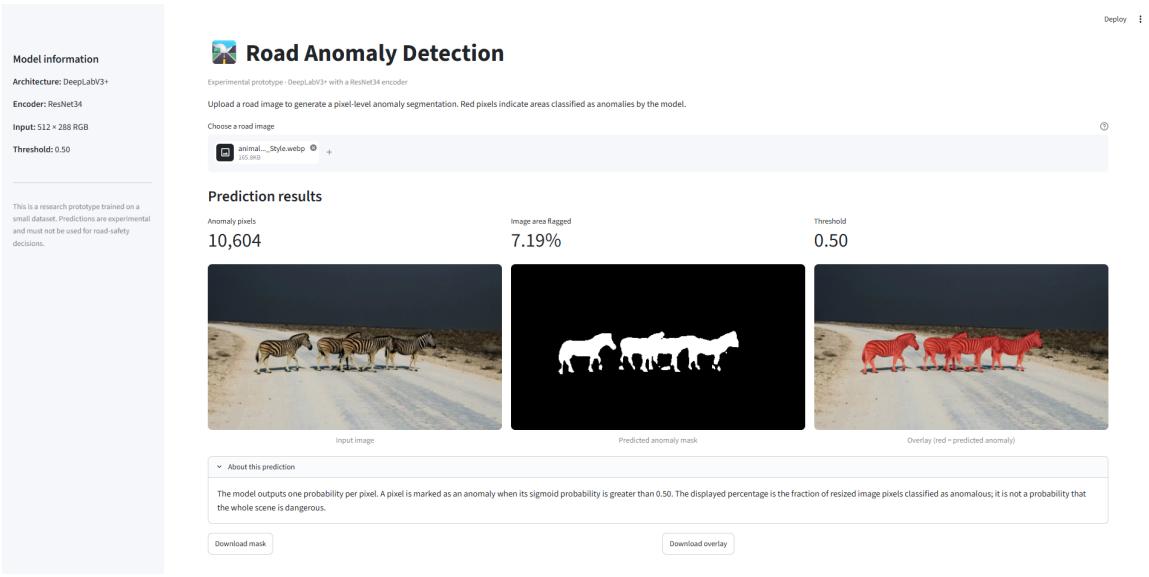
The application is a web interface built with Streamlit. It implements the segmentation objective (DSO2): the user uploads a road image and the application:

1. resizes the image to 512×288 and normalizes it with the ImageNet mean and standard deviation, exactly as during training;
2. runs the DeepLabV3+ model, which outputs one probability per pixel (sigmoid);
3. thresholds the probabilities at 0.50 to obtain the binary anomaly mask;
4. displays the input image, the predicted mask and an overlay where the anomalous pixels are shown in red, together with the number of anomalous pixels and the percentage of the image flagged;
5. lets the user download the mask and the overlay.

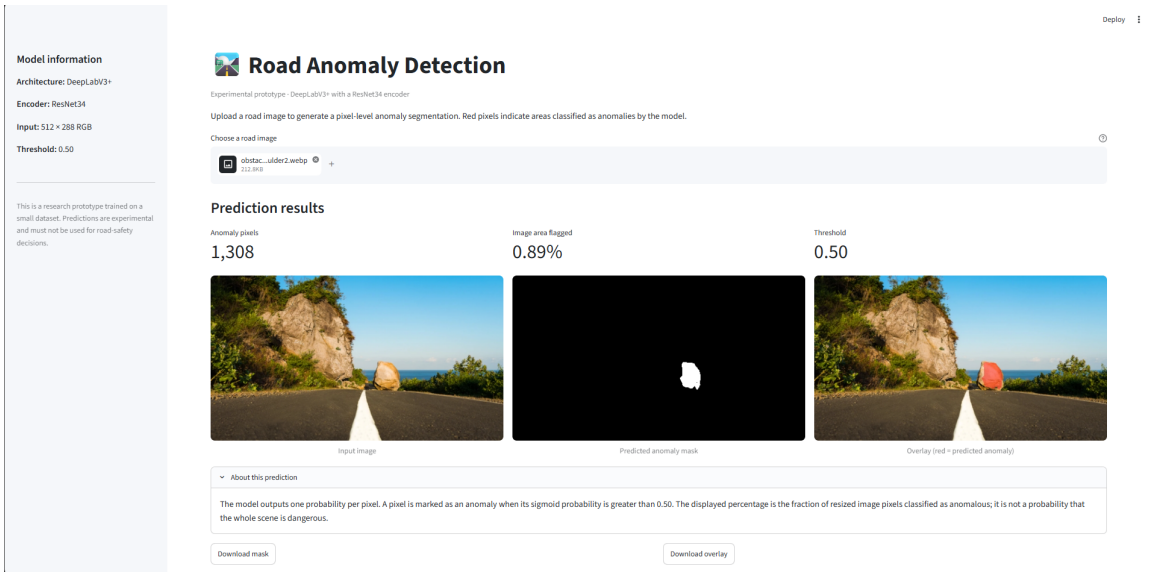
The sidebar shows the model information (architecture, encoder, input size, threshold) and a warning that the prototype is experimental and must not be used for road-safety decisions. The displayed percentage is the fraction of flagged pixels of the resized image; it is not a probability that the scene is dangerous. The image classifier of DSO1 is not integrated into this prototype.

7.4 Demonstration

Figures 7.1 and 7.2 show the application on four images of the RoadAnomaly dataset, one for each of the main anomaly types: animals, cones, an obstacle (a boulder) and a construction scene with a vehicle. Since the application does not display the ground truth, the figures report the number and percentage of flagged pixels, not Dice scores.

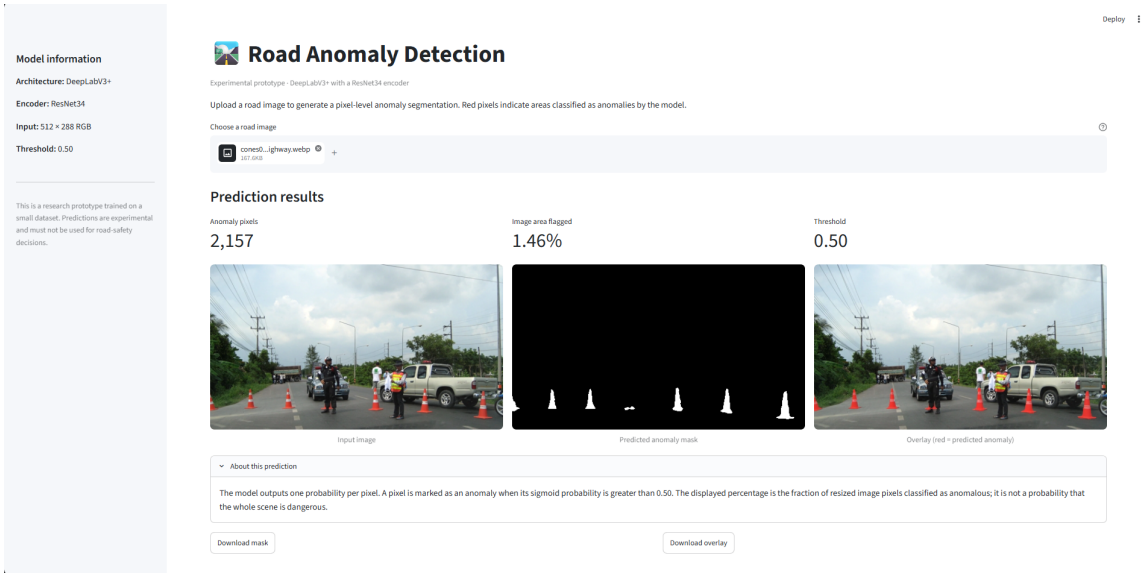


(a) Animals: four zebras crossing the road (10,604 anomalous pixels, 7.19% of the image).

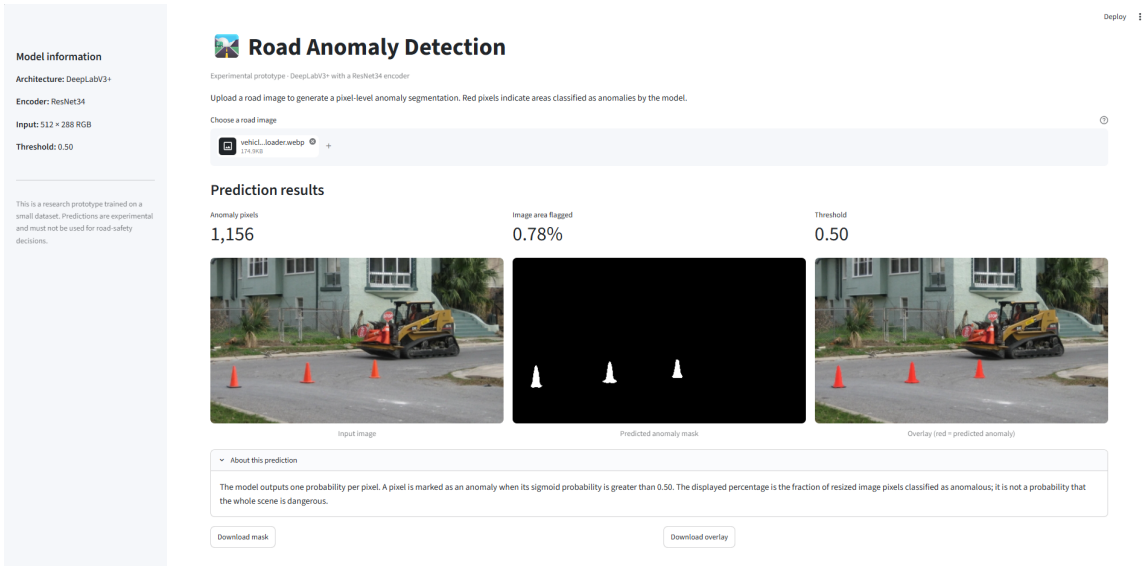


(b) Traffic cones at a highway checkpoint (2,157 anomalous pixels, 1.46%).

Figure 7.1: Screenshots of the prototype on an animal image and a cone image.



(a) Obstacle: a boulder on the road (1,308 anomalous pixels, 0.89%).



(b) Construction scene with a loader and cones (1,156 anomalous pixels, 0.78%).

Figure 7.2: Screenshots of the prototype on an obstacle image and a construction-scene image.

The four examples illustrate the behavior of the model:

- **Animals:** the four zebras are segmented with outlines that follow their bodies closely, and the road is not flagged. Adjacent animals merge into a single region, and thin parts such as the legs are less precise.
- **Cones:** the cones are detected as separate regions, including small distant ones, while the people and vehicles of the scene are correctly left unflagged.
- **Obstacle:** the boulder lying on the road is segmented as a compact region. The large rock face on the left side of the image is not flagged, so the model distinguishes the obstacle on the road from the surrounding scenery.

- **Construction scene:** the three cones are flagged, but the loader is not. The file name of this image indicates the vehicle category, so the loader is probably the annotated anomaly. If so, this is a missed detection, consistent with the low recall observed in the evaluation and with the fact that the vehicle category has only three images in the dataset.

7.5 Potential practical use

- **Annotation assistance:** pre-labeling anomalies in new road images, to be corrected by a person.
- **Road inspection:** flagging survey images that contain obstacles (animals, rocks, lost tires) for review.
- **Starting point for driver-assistance research,** to be improved with more data and open-set methods.

7.6 Limits and improvements

The deployed model was trained on only 34 images and inherits the limits described in Section 6.5: it can miss small or scattered objects and unusual shapes, and it only recognizes the types of anomalies seen during training. We did not measure inference speed or test video, so the prototype is not suitable for real-time or safety-critical use. Predictions in the application may differ from the per-image scores in the evaluation, since results vary between training runs on such a small dataset.

Possible improvements are a threshold slider to favor recall, an ensemble of U-Net and DeepLabV3+, the integration of the anomaly-category classifier (DSO1) next to the segmentation mask, video input, and a lighter model for embedded devices.

7.7 Conclusion

A prototype web application based on DeepLabV3+ turns the segmentation model into a tool that produces an anomaly mask, an overlay and simple statistics from an uploaded road image. The examples show that animals, cones and boulders are localized well, while vehicles can still be missed. The prototype is a proof of concept for annotation assistance and road inspection, not a component for real-time or safety-critical use.

General Conclusion

This project addressed the road safety of autonomous vehicles by investigating road anomaly detection on the RoadAnomaly dataset, following the CRISP-DM methodology. The work covered problem definition, dataset exploration, preprocessing, model development, quantitative and qualitative evaluation, and deployment considerations.

The global objective was broken down into two Data Science objectives. For DSO1, image classification, a small CNN trained from scratch was compared with a ResNet34 pretrained on ImageNet to recognize the anomaly category (animals, obstacles or other). For DSO2, pixel-level segmentation, four approaches were compared: a U-Net trained from scratch, a U-Net with a pretrained ResNet34 encoder, DeepLabV3+ with a pretrained ResNet34 encoder, and a convolutional autoencoder.

For segmentation, the results show that transfer learning is the most effective strategy among the tested approaches. In 5-fold grouped cross-validation, U-Net+ResNet34 reaches a Dice score of 0.789 ± 0.094, while DeepLabV3+ reaches 0.798 ± 0.061. Their observed average performances are close. DeepLabV3+ obtains the best Dice on the single held-out test split, with a score of 0.764. The scratch U-Net obtains 0.589, while the autoencoder obtains 0.207. For classification, the pretrained ResNet34 performs better than the small CNN in cross-validation, with a mean accuracy of 0.784 against 0.435.

The experiments also reveal important limitations. The segmentation models struggle with small scattered rocks and scenes that are not represented in training. The autoencoder does not provide competitive segmentation performance in its current configuration. The small number of examples limits the reliability of the classification results, and the dataset contains only anomalous images, so the classifiers recognize the type of anomaly rather than its presence.

Overall, the project demonstrates the usefulness of CNNs and transfer learning for both road anomaly classification and segmentation, while highlighting the difficulty of generalizing from a very small dataset. Future work should focus on collecting more diverse data, including images without anomalies, preserving small-object details, improving recall and investigating methods designed specifically for open-set anomaly detection. The current solution is an academic proof of concept; additional robustness, latency and safety validation would be required before practical deployment.

List of Abbreviations

Acc.	Accuracy
ASPP	Atrous Spatial Pyramid Pooling
AuPRC	Area under the Precision–Recall Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
BCE	Binary Cross-Entropy
CNN	Convolutional Neural Network
CRISP-DM	Cross-Industry Standard Process for Data Mining
FN	False Negative
FP	False Positive
FPR95	False Positive Rate at 95% True Positive Rate
IoU	Intersection over Union
MSE	Mean Squared Error
R34	ResNet34
ResNet	Residual Network
RGB	Red, Green, Blue
Std	Standard Deviation
TP	True Positive